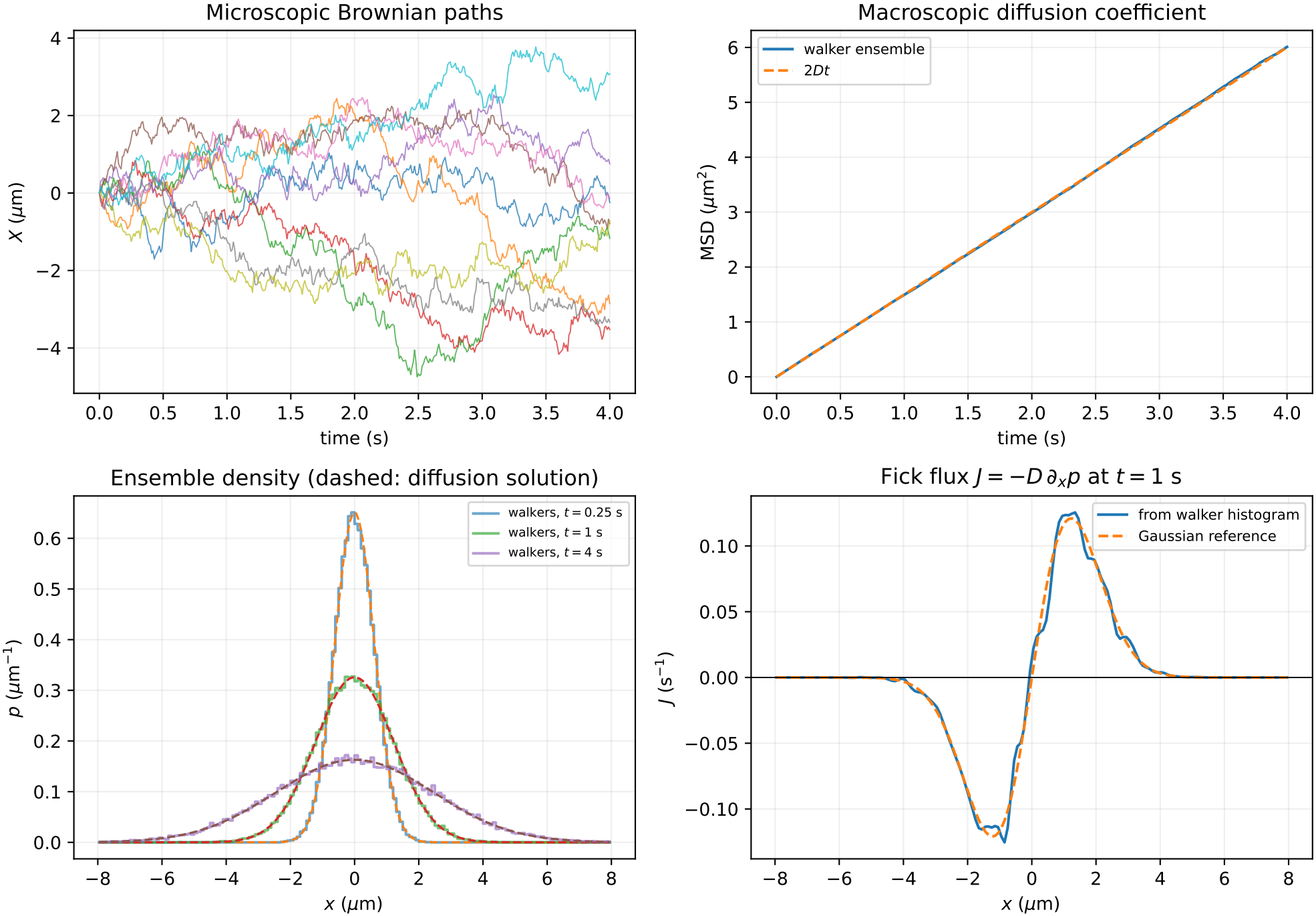

Microscopic Brownian paths
X (μm)
time (s)
Macroscopic diffusion coefficient
walker ensemble
2Dt
MSD (μm²)
time (s)
Ensemble density (dashed: diffusion solution)
walkers, t = 0.25 s
walkers, t = 1 s
walkers, t = 4 s
p (μm⁻¹)
x (μm)
Fick flux J = −D ∂ₓp at t = 1 s
from walker histogram
Gaussian reference
J (s⁻¹)
x (μm)